

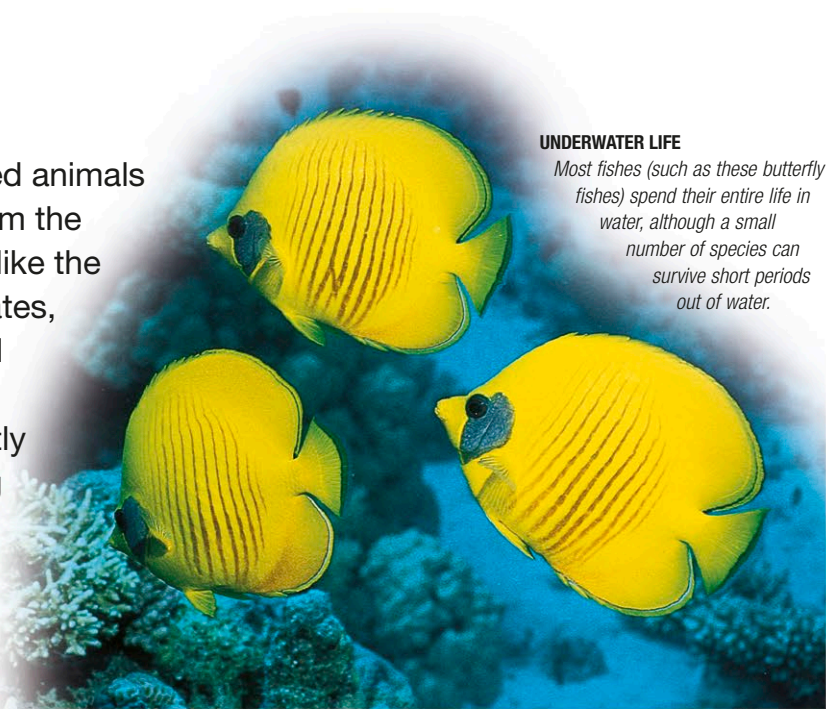
FISHES

PHYLUM	Chordata
CLASSES	Cephalaspidomorphi, Chondrichthyes, Osteichthyes
ORDERS	83
FAMILIES	About 540
SPECIES	About 33,000

Fishes were the first backboned animals to appear on Earth, and they form the largest group of vertebrates. Unlike the other major divisions of vertebrates, however, fishes are not a natural group; instead, they are an

informal collection of 8 classes that are only distantly related to one another. A typical fish breathes using gills, has a body covered with scales, maneuvers using fins, and is ectothermic (cold-blooded). Most species live in either freshwater or the sea, but a few species move between both environments.

UNDERWATER LIFE
Most fishes (such as these butterfly fishes) spend their entire life in water, although a small number of species can survive short periods out of water.



Evolution

Living fishes are usually divided into 3 major groups: jawless, cartilaginous, and ray-finned. Each of these groups had a different ancestor and evolved separately from the others.

The first fishes appeared more than 500 million years ago, having probably evolved from soft-bodied, filter-feeding invertebrates. Like living jawless fishes, these fishes had a round, fleshy mouth without jaws, but unlike them, many lacked teeth.

The first fishes (indeed, the first animals) with movable jaws arose around 440 million years ago. Known as acanthodians (or spiny sharks), their jaws evolved from their front gill arches. Linked to the development

of jaws was the appearance of teeth. Having jaws and teeth made it possible for acanthodians to feed on a wide range of food. They also had several pairs of spines along the lower sides of their body, from which the paired fins of fishes evolved. Unlike the early jawless fishes, their body was covered with scales.

Fishes with a cartilage skeleton first appeared about 370 million years ago. These were the ancestors of living sharks, skates, and rays.

Some 50 million years before the appearance of cartilaginous fishes, a group of fishes with a bony internal skeleton had appeared. There are 2 main groups of fishes: fleshy-finned fishes, which have fins supported by bony skeletons fringed with rays; and ray-finned fishes, in which the fins are supported only by fin rays. The fleshy-finned fishes almost certainly gave rise to the first 4-limbed land vertebrates. The ray-finned fishes appeared at about the same time but have since become a far larger and more successful group. They gradually developed features that made them more efficient swimmers, including thinner scales, highly flexible

fins, and a symmetrical tail. Also important were the evolution of more mobile jaws and gill chambers that could be more easily ventilated.

Anatomy

Fishes have several physical adaptations for life in water. The body of a fish is covered with smooth scales, and has fins to provide power, steering, and stability. All fishes have gills for extracting oxygen from water, and all have an internal

skeleton, although this takes different forms in each of the 3 main groups of fishes. In jawless fishes, the body is supported by a simple rod called the notochord and some have rudimentary vertebrae. Sharks, skates, and rays

FIN TYPES
A fish's dorsal, anal, and tail fins occur singly along the midline of the body, while the pectoral and pelvic fins are paired. The common carp (shown here) has only one dorsal fin; however, some other fish have 2 or 3 dorsal fins.

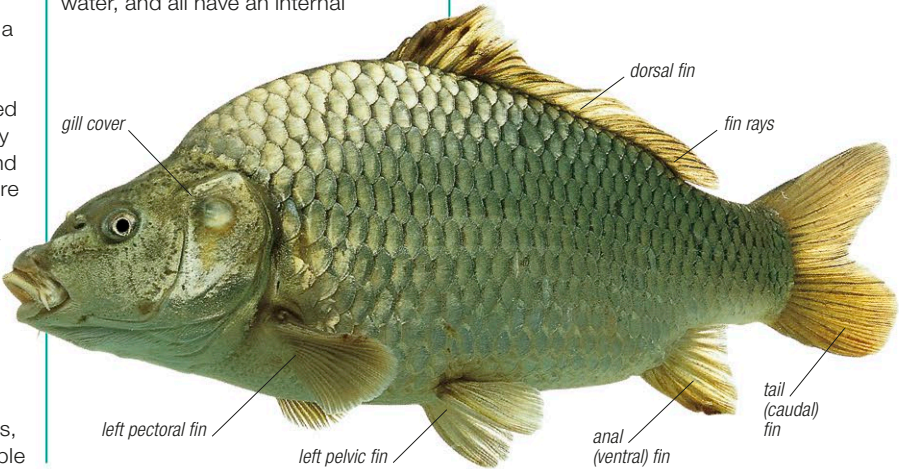


CEPHALASPIS



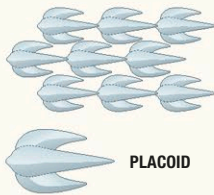
STICHOCENTRUS

FISH FOSSILS
Among the first fishes to appear was a form called *Cephalaspis* (top). It had a jawless mouth, and its head and gills were protected by a large, bony shield. By contrast, a fossil of the bony fish *Stichocentrus* (above) reveals jaws, scales, and a much lighter and more flexible internal skeleton.

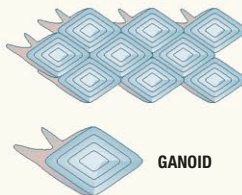


SCALES

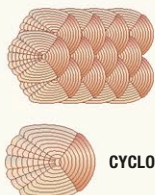
Fishes have several types of scales, made of various materials. Cartilaginous fishes have toothlike placoid scales (dermal denticles). Some primitive bony fishes have thick, relatively inflexible scales. They include diamond-shaped ganoid scales, found in sturgeons, and layered cosmoid scales, seen in coelacanth. Like placoid scales, these are made of dentine and a substance similar to enamel. Bony fishes have thin scales made of bone, with one end embedded in the skin and the other exposed: cycloid scales have a smooth exposed surface, while ctenoid scales are rough or spiny.



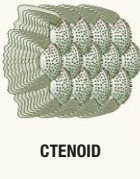
PLACOID



GANOID



CYCLOID



CTENOID

SCALE PATTERNS
Placoid scales are embedded in the skin, with the widest part facing outward, and give the skin a rough texture. Ganoid scales interlock and are joined to one another by fibres. Cycloid and ctenoid scales are arranged so that the exposed surfaces overlap with one another, forming a smooth, flexible covering.